

Exercise 6 – Evaluation II

Introduction to Machine Learning

Hint: Useful libraries

R

```
# Consider the following libraries for this exercise sheet:
```

```
library(mlbench)
library(mlr3)
library(mlr3learners)
```

Python

```
# Consider the following libraries for this exercise sheet:
```

```
# general
import numpy as np
import pandas as pd
```

```
# sklearn
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.model_selection import RepeatedKFold
from sklearn.model_selection import RepeatedStratifiedKFold
```

Exercise 1: Overfitting & underfitting

Learning goals

1. Discuss for given situations if they might lead to overfitting / underfitting
2. Judge whether over- and underfitting are properties of a learner or of a fixed model
3. Discuss to what extent over- and underfitting can be avoided

Assume a polynomial regression model with a continuous target variable y , a continuous, p -dimensional feature vector $\mathbf{x}$ and polynomials of degree d , i.e.,

$$f(\mathbf{x}^{(i)}) = \theta_0 + \sum_{j=1}^p \sum_{k=1}^d \theta_{j,k} (\mathbf{x}_j^{(i)})^k.$$

- a) For each of the following situations, indicate whether we would generally expect the performance of a flexible polynomial learner (high d) to be better or worse than an inflexible one (low d). Justify your answer.

NB

We can only state tendencies here; performance strongly depends on the specific data situation.

- i. The sample size n is extremely large, and the number of features p is small.
 - ii. The number of features p is extremely large, and the number of observations n is small.
 - iii. The true relationship between the features and the response is highly non-linear.
 - iv. The data could only be observed with a high level of noise.
- b) Are overfitting and underfitting properties of a learner or of a fixed model? Explain your answer.
- c) Should we aim to completely avoid both overfitting and underfitting?

Exercise 2: Resampling in practice

Learning goals

1. Implement resampling procedures in R/Python
2. Understand how the choice of resampling strategy affects the quality of the GE estimator

- a) Why would we apply resampling rather than a single holdout split?
- b) Classify the `german_credit` data into solvent and insolvent debtors using logistic regression. Compute the training error w.r.t. MCE.

Python Hint

Read the already preprocessed file [german_credit_for_py.csv](#)

- c) In order to evaluate your learner, compare the test MCE using
 1. three times ten-fold cross validation (3x10-CV)
 2. 10x3-CV
 3. 3x10-CV with stratification for the feature `foreign_worker` to ensure equal representation in all folds
 4. a single holdout split with 90% training data

Hint

R

You will need `rsmp`, `resample` and `aggregate`.

Python

You will need `RepeatedKFold`, `RepeatedStratifiedKFold` and `train_test_split`.

- d) Discuss and compare your findings and compare them to the training error computed previously.

Exercise 3: Computational cost of resampling

The whole exercise is only for lecture group B!

Learning goals

1. Determine how many models a resampling strategy fits
2. Reason about the computational cost of different resampling strategies
3. Relate computational cost to the quality of the GE estimate

Recall that a resampling strategy is a collection of B train/test splits,

$$\mathcal{J} = ((J_{\text{train},1}, J_{\text{test},1}), \dots, (J_{\text{train},B}, J_{\text{test},B})),$$

and that the learner is refitted on each of the B training sets.

- a) A data set has $n = 200$ observations. State how many models are fitted for each of the following strategies:
- i. 10-fold cross-validation (10-CV)
 - ii. 5-times repeated 10-fold cross-validation (5x10-CV)
 - iii. leave-one-out CV (LOO-CV)
 - iv. subsampling with 25 iterations

Hint

The number of model fits equals the number of splits B .

- b) Now give the number of model fits **in general**, i.e. as a formula, for a data set of size n with k folds, r repetitions and B iterations:
- i. a single holdout split
 - ii. k -fold CV
 - iii. r -times repeated k -fold CV
 - iv. LOO-CV
 - v. subsampling or bootstrap with B iterations
- c) Explain why LOO-CV becomes computationally prohibitive for large n , and name a resampling strategy that achieves a similarly low (pessimistic) bias at much lower cost.
- d) Both 3-times repeated 10-fold CV (3x10-CV) and 10-times repeated 3-fold CV (10x3-CV) fit exactly 30 models, so their computational cost is roughly the same. Does this mean that both strategies provide GE estimates of the same quality? Explain your answer.

Exercise 4: Variance and independence of GE estimators

The whole exercise is only for lecture group A!

Learning goals

1. Derive mean and variance of the holdout GE estimator on a fresh i.i.d. test set
2. Understand why the per-fold errors of cross-validation are not independent

Consider a fixed model $\hat{f}$ that has been fitted on some training data and is now held fixed. We estimate its generalization error on a test set $\mathcal{D}_{\text{test}}$ of n_{test} observations drawn i.i.d. from $\mathbb{P}_{xy}$, independently of the training data:

$$\widehat{\text{GE}}(\hat{f}) = \frac{1}{n_{\text{test}}} \sum_{(\mathbf{x}, y) \in \mathcal{D}_{\text{test}}} L(y, \hat{f}(\mathbf{x})).$$

- a) Show that $\widehat{\text{GE}}(\hat{f})$ is an unbiased estimator of the theoretical generalization error $\text{GE}(\hat{f}) = \mathbb{E}[L(y, \hat{f}(\mathbf{x}))]$ of the fixed model $\hat{f}$.
- b) Show that $\text{Var}[\widehat{\text{GE}}(\hat{f})] = \frac{1}{n_{\text{test}}} \text{Var}[L(y, \hat{f}(\mathbf{x}))]$ and interpret how the estimator's precision depends on the test set size.
- c) In k -fold cross-validation we obtain k per-fold estimates $\widehat{\text{GE}}^{(1)}, \dots, \widehat{\text{GE}}^{(k)}$ and report their mean. One might be tempted to treat these k numbers as k independent measurements – e.g., to report their standard deviation or run a t -test to compare two learners. Explain why the $\widehat{\text{GE}}^{(j)}$ are **not** independent, and what this implies for such a variance estimate.